

Querying SharePoint Data in Power BI

Options & Performance

March 4th, 2026



About Me

Experience

- 12+ years with Microsoft BI stack
- 10+ years with Power BI
- 10+ years with SharePoint

Credentials

- MCSE: Productivity
- MS: PBI Data Analyst
- PMP



John Kerski



Contact Me

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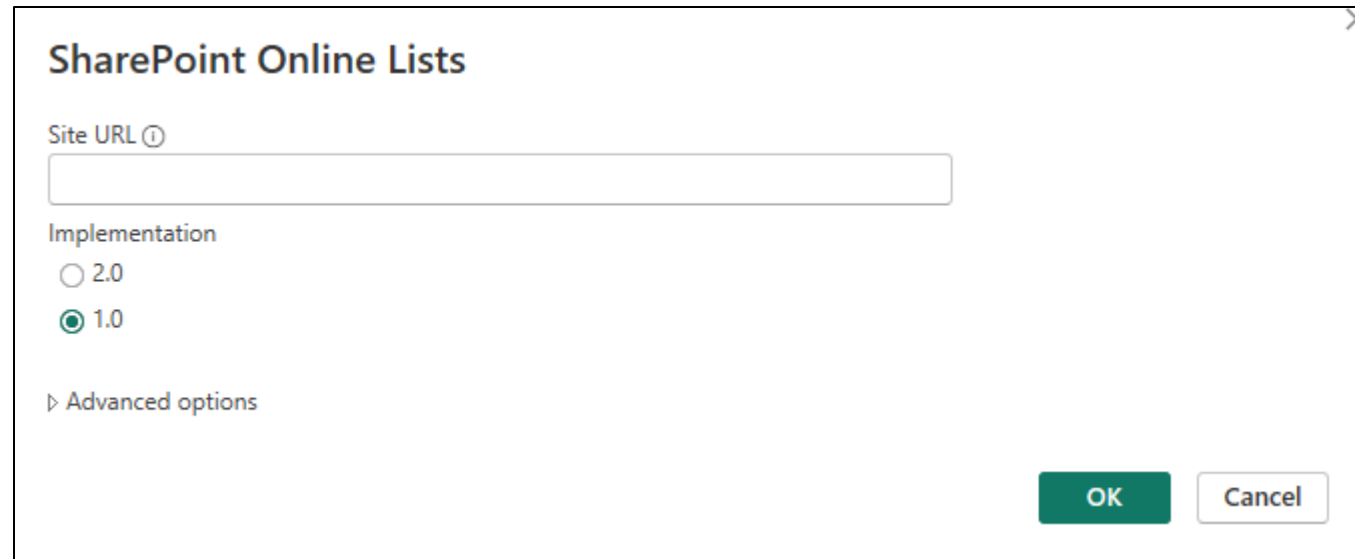


Agenda

- Lists
 - Options
 - Performance
- Document Libraries
 - Options
 - Performance

SharePoint Lists

SharePoint Online List Version 1



A screenshot of a configuration dialog box titled "SharePoint Online Lists". The dialog has a close button (X) in the top right corner. It contains a text input field for "Site URL" with a help icon (i) to its right. Below this is a section labeled "Implementation" with two radio button options: "2.0" and "1.0". The "1.0" option is selected, indicated by a green dot. At the bottom left, there is a link "Advanced options" preceded by a right-pointing triangle. At the bottom right, there are two buttons: "OK" (green) and "Cancel" (white with a grey border).

SharePoint Online Lists

Site URL ⓘ

Implementation

☐ 2.0

☒ 1.0

▸ Advanced options

OK Cancel

**SharePoint
V1 Connector**

Power BI



With larger lists...



V1 - Pros & Cons

Pros

- Easy To Use
- Good for small lists (< 5,000 items)

Cons

- Get More Data Than Needed
- Lots of “Cleanup” Power Query Steps
 - Remove Columns
 - Expand Columns
- Slow for larger lists
 - Especially with several lookups

Roche's Maxim on Data Transformation

Data should be transformed as far upstream as possible, and as far downstream as necessary

SharePoint Online List Version 2

×

SharePoint Online Lists

Site URL ⓘ

Implementation
☒ 2.0
☐ 1.0

⌵ Advanced options
View Mode

Default - Retrieve the columns set in... ▾

OK

Cancel

V2 - Pros & Cons

Pros

- Easy To Use
- Good for small lists (< 5,000 items)

Cons

- Public View dependency could lead to schema drift
 - Or Errors
- Slow for larger lists
 - Especially with several lookups

OData

×

OData feed

☒ Basic ☐ Advanced

URL

OData - Example

ABC 123	firstname	ABC 123	gender	ABC 123	sentence	ABC 123	bool	ABC 123	ID	ABC 123	Modified	ABC 123	Modified By
	Delbert		Male		Cum sociis natoque penatibus et magnis dis parturient montes, nascet...		TRUE		1		2024-02-27T20:54:03Z		Record
	Nell		Female		Curabitur in libero ut massa volutpat convallis. Morbi odio odio, eleme...		TRUE		2		2024-02-27T20:54:03Z		Record
	Khalil		Male		Morbi vestibulum, velit id pretium iaculis, diam erat fermentum justo, ...		TRUE		3		2024-02-27T20:54:03Z		Record
	Gordan		Male		Morbi vel lectus in quam fringilla rhoncus. Mauris enim leo, rhoncus se...		TRUE		4		2024-02-27T20:54:03Z		Record
	Hayyim		Male		Morbi a ipsum. Integer a nibh. In quis justo.		FALSE		5		2024-02-27T20:54:03Z		Record
	Jordan		Male		Donec quis orci eget orci vehicula condimentum.		FALSE		6		2024-02-27T20:54:04Z		Record
	Currey		Male		In est risus, auctor sed, tristique in, tempus sit amet, sem. Fusce conse...		FALSE		7		2024-02-27T20:54:04Z		Record
	Deane		Male		Suspendisse potenti. Nullam porttitor lacus at turpis. Donec posuere ...		TRUE		8		2024-02-27T20:54:04Z		Record
	Tracy		Female		Nulla ut erat id mauris vulputate elementum. Nullam varius. Nulla facil...		FALSE		9		2024-02-27T20:54:04Z		Record
	Tobiah		Male		Nam congue, risus semper porta volutpat, quam pede lobortis ligula, s...		FALSE		10		2024-02-27T20:54:04Z		Record

`/_api/lists/GetByTitle('SampleData')/items?$select=field_1,field_3,field_6,field_7,ID,Modified,Editor/EMail,Editor/Title&$expand=Editor&$top=5000`

OData - Pros & Cons

Pros

- Good for large lists

Cons

- You must learn OData syntax:
[Working with lists and list items with REST | Microsoft Learn](#)
- Defaults to SharePoint Paging Limit
- 100-page limit
 - 5,000 items = 50 network calls
- Credentials required for each call

Custom-made (Web.Contents)

fnGetFields

- *You*
 - Supply list name
- *Function*
 - Contacts the /fields endpoint in SharePoint to retrieve the pertinent columns about the SharePoint list.

Table of Fields

- *You*
 - Filter fields furthermore to refine query.

fnGetSharePointData

- *You*
 - Supply the Table of Fields
 - Provide any additional filters.
- *Function*
 - Converts Fields into Odata query.
 - Automatically expand results.

[An alternative way to import SharePoint list data into Power BI. \(github.com\)](#)

[Share a query - Power Query | Microsoft Learn](#)



Results on a Fabric Capacity (> 50,000 items)

Dataset	Median – Seconds to Refresh	Standard Deviation – Seconds to Refresh	Capacity Units	Number of Refreshes
V2	151.50	41.98	5,238.66	30
V1	103.20	8.36	5,053.90	30
OData	37.47	7.22	3,197.31	30
Custom- made	18.93	2.81	1,151.65	30

Custom-made - Pros & Cons

Pros

- Good for large lists
- You don't have to learn OData syntax
- You can use template to generate OData for you

Cons

- Combining Data from different SharePoint site may conflict with privacy settings
 - Alternative: Do this in dataflows.

SharePoint Lists – Best Practices

- Avoid V2
- New to Power Query
 - Consider OData for Large Lists
- Experienced with Power Query
 - Switch to Custom-made approach

SharePoint Document Libraries

SharePoint Folder (a.k.a. SharePoint.Files)

×

SharePoint folder

Site URL ⓘ

ABC ▾

OK

Cancel

You want SharePoint files?



SharePoint.Files - Pros & Cons

Pros

- Best suited for:
 - Documents of a single, consistent type.
 - Site has one Document Library
 - Combining files across Document Libraries

Cons

- Slower developer experience
 - Loads all file metadata
- Slower when > 100 files

SharePoint.Contents

✕ ✓ <i>fx</i> = SharePoint.Contents(SharePoint_URL)			
	ABC123 Content	ABC Name	ABC Extension
1	Table	Appfiles	
2	Table	Documents	
3	Table	FormTemplates	
4	Table	ListTemplateGallery	
5	Table	MasterPageGallery	
6	Table	SiteAssets	
7	Table	SitePages	
8	Table	SolutionGallery	

SharePoint.Contents - Pros & Cons

Pros

- Best suited for:
 - Sites with lots of files
 - Files in specific folders

Cons

- Need multiple tables if pulling multiple document libraries.

OneLake Shortcuts – SharePoint Folder

New shortcut

lh_onedrive_shortcuts is located in the region **East US 2**. Any data sourced through this shortcut will be processed in the same region.

S

SharePoint Folder
(Preview)
File
[Learn more](#)

☒ Existing connection ☐ New connection

Connection:

Select connection

Refresh

Create shortcuts directly in your folders
When specifying the location, enter a site or Teams URL in one of these formats:

- Default site: https://{tenant-name}.sharepoint.com/
- Site: https://{tenant-name}.sharepoint.com/Sites/{Site-Name}
- Team: https://{tenant-name}.sharepoint.com/Teams/{Team-Name}

Previous

Next

Cancel

Lakehouse

Data Connectivity mode ⓘ
☒ Import ☐ DirectQuery

OK

Cancel

OneLake Shortcuts - Pros & Cons

Pros

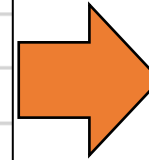
- No Manual Pipelines
- Updated every 2 minutes
- Allows you to process outside of the semantic model

Cons

- In Preview (as of Jan. 2026)
- Need Fabric Licensing
 - Unsupported in GCC/DoD
- Slower if not processing outside of semantic model

SharePoint.Contents - Incremental

= SharePoint.Contents(SharePoint_URL)			
	ABC 123 Content	ABC Name	ABC Extension
1	Table	Appfiles	
2	Table	Documents	
3	Table	FormTemplates	
4	Table	ListTemplateGallery	
5	Table	MasterPageGallery	
6	Table	SiteAssets	
7	Table	SitePages	
8	Table	SolutionGallery	



Incremental refresh and real-time data

Refresh large tables faster with incremental refresh. Plus, get the latest data in real time with DirectQuery (Premium only). [Learn more](#)

These settings will apply when you publish the dataset to the Power BI service. Once you do that, you won't be able to download it back to Power BI Desktop. [Learn more](#)

- Select table**

Test Import 1 & 3
- Set import and refresh ranges**

☒ Incrementally refresh this table

Archive data starting

1

Years

 before refresh date

Data imported from 1/1/2024 to 12/22/2025 (inclusive)

Incrementally refresh data starting

1

Days

 before refresh date

Data will be incrementally refreshed from 12/23/2025 to 12/23/2025 (inclusive)
- Choose optional settings**

☐ Get the latest data in real time with DirectQuery (Premium only) [Learn more](#)

Selected table cannot be folded for DirectQuery.

☐ Only refresh complete day [Learn more](#)

☐ Detect data changes [Learn more](#)

Apply

Cancel

Results on a Fabric Capacity (> 1,000 Files)

Dataset	Median – Seconds to Refresh	Standard Deviation – Seconds to Refresh	Capacity Units	Number of Refreshes
One-Lake Shortcuts	568.2	154.9	5,139.57	10
SharePoint.Contents	461.9	173.0	2,613.52	10
SharePoint.Files	408.0	182.3	3,266.31	10
SharePoint.Contents - Incremental	68.6	155.4	204.57	10

SharePoint.Contents - Incremental - Pros & Cons

Pros

- Faster Reloads

Cons

- More Setup Steps

Document Libraries – Best Practices

- Use SharePoint.Contents for Large Libraries
- Implement Incremental
- Add Development Parameter
- Future
 - Consider Shortcuts

Thank you!

